

Does Compression Forget Cancer?

Dangerous-Class Degradation in Medical Vision Transformer Pruning:
A Multi-Seed Clinical Evaluation of Pruning, Quantization, and Distillation

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TinyML / Efficient Deep Learning Final Project

April 2026

Abstract

Compressing medical vision models for edge deployment is usually evaluated with aggregate accuracy, yet in class-imbalanced clinical datasets aggregate metrics can hide catastrophic degradation on rare, dangerous classes. We fine-tune DeiT-Tiny and DeiT-Small on the HAM10000 dermoscopy benchmark and evaluate three families of compression—**pruning and sparsity**, **post-training quantization**, and **knowledge distillation**—through the lens of a newly introduced **Dangerous-Class Degradation Ratio (DCR)** together with per-class AUROC, expected calibration error (ECE), per-layer class-conditional gradient flow, and quantitative attention-mask overlap. All headline results are reported across $n=4$ random seeds at the cell level and $n=3$ for paired significance tests, with lesion-grouped patient splits.

Our results reshape the narrative of compression in medical vision. *Magnitude pruning*, the default baseline in most compression papers, exhibits the most dangerous failure mode: its DCR reaches 2.15 on DeiT-Small, meaning dangerous classes (*mel*, *bcc*, *akiec*) degrade at more than twice the rate of safer classes. *Taylor (first-order) pruning*, despite its simplicity, emerges as the Pareto-optimal clinical choice: highest melanoma AUROC (0.928), best calibration ($\text{ECE} = 0.033$), and DCR below 1 at 50% sparsity. *Wanda* (activation-aware) pruning preserves minority-class sensitivity through *classifier-head prediction collapse* ($3\times$ ECE degradation, AUROC below magnitude’s), and—contrary to our earlier qualitative reading—its attention-mask IoU and pointing-game accuracy across $n=5$ seeds *exceed* magnitude’s and match the dense baseline; the failure is downstream of attention, not in feature localization. We isolate the underlying mechanism: per-layer pruning rate is positively correlated with per-class gradient mass for magnitude ($r \approx 0.62\text{--}0.74$, $p < 10^{-5}$), uncorrelated for Wanda ($|r| < 0.20$), and *negatively* correlated for Taylor ($r \approx -0.41$). Three distinct failure modes follow—magnitude’s gradient-coupling, Wanda’s head-collapse, Taylor’s deliberate gradient-preservation—and an earlier activation-outlier hypothesis was disconfirmed and is preserved for transparency. *INT8 quantization* is the safest single compression lever ($3.8\times$ size reduction, no measurable clinical degradation), and *knowledge distillation* improves dense performance but does not produce pruning-robust features. We conclude that compression safety in medical AI cannot be inferred from balanced accuracy or melanoma sensitivity alone, and recommend the DCR + AUROC + ECE triplet, accompanied by per-layer gradient-coupling and attention-mask overlap, as a minimal clinical evaluation protocol.

1 Introduction

Dermatology is one of the clearest practical targets for edge-deployed medical AI. Portable, offline skin-lesion classifiers could support screening in low-connectivity settings, reduce latency, and preserve patient privacy by avoiding cloud inference. All of these benefits require *compact* models that fit within the memory, compute, and energy budgets of clinic-side hardware. Compression is therefore not optional in this domain; it is the prerequisite for deployment.

The problem is that the standard evaluation of compressed models—top-1 or balanced accuracy on a held-out validation set—can be profoundly misleading in clinical settings. HAM10000, the most widely used dermoscopy benchmark, is strongly imbalanced: melanocytic nevi dominate the training set

while the deadliest class, melanoma, is comparatively rare. A compressed model can therefore preserve aggregate accuracy *while losing the ability to detect exactly the diseases clinicians care most about*. This asymmetric failure—which we term **dangerous-class degradation**—is the central concern of this paper.

We ask: *when a medical Vision Transformer is compressed, does it forget dangerous diseases first, and does the choice of compression method determine whether it does?* To answer this, we conduct a multi-seed, patient-split, clinically evaluated comparison of three compression families—pruning, quantization, and knowledge distillation—using metrics that expose class-wise safety rather than obscure it.

Contributions.

1. We introduce the **Dangerous-Class Degradation Ratio (DCR)** as a single-number safety metric for class-imbalanced compression evaluation, computed as the ratio of mean sensitivity drop on clinically dangerous classes to mean drop on safer classes.
2. We show that **magnitude pruning**, the common default baseline, has $\text{DCR} \approx 2.15$ on DeiT-Small—more than twice the expected symmetric degradation—and is therefore the *most clinically dangerous* pruning criterion in our study despite its competitive balanced accuracy.
3. We identify **Taylor pruning** as Pareto-optimal across all clinical metrics at 50% sparsity, challenging the common assumption that simple first-order methods are outperformed by more sophisticated activation-aware approaches in medical vision.
4. We show that **Wanda**’s apparent performance on minority classes reflects *prediction collapse* at the classifier head, not improved discrimination: calibration degrades by $\sim 3\times$ (ECE 0.099 vs. 0.035) while melanoma AUROC drops below magnitude’s.
5. We identify the **mechanism behind magnitude’s class-asymmetric damage**: per-layer pruning rate is positively correlated with per-layer gradient mass for both dangerous and safer classes (Pearson $r \approx 0.62$ and 0.74 respectively, $p < 10^{-5}$ across all five seeds). Wanda is uncorrelated with gradient mass ($|r| < 0.20$); Taylor is negatively correlated ($r \approx -0.41$). This isolates three distinct failure modes: magnitude’s gradient-coupling, Wanda’s head-collapse (with attention preserved), and Taylor’s deliberate gradient-preservation. We also report that an earlier activation-outlier hypothesis was disconfirmed ($|r| < 0.11$); the disconfirmed result is preserved in §4.4 for transparency.
6. We provide **quantitative attention-mask overlap** (IoU and pointing-game) using the Tschandl HAM10000 lesion segmentations across $n=5$ seeds. Wanda’s attention IoU (0.43 ± 0.02) and pointing-game accuracy ($91.4\% \pm 0.8\%$) *match or exceed* those of the dense model and substantially exceed magnitude’s ($0.41, 84.7\%$). Wanda’s failure is not attention misallocation; it is downstream classifier collapse.
7. We show that **INT8 quantization** is the safest compression stage (size $\div 3.8$, no measurable clinical degradation), while **knowledge distillation** improves dense performance but *does not* transfer pruning-robustness to the student.

2 Background and Related Work

Pruning and sparsity. Magnitude pruning [1] remains the field’s default baseline for its simplicity and surprising robustness. First-order Taylor scoring [2] estimates the effect of removing a weight via the product of weight and gradient, implicitly incorporating task information. Wanda [3] proposed an activation-aware score $|w| \cdot \|x\|_2$ for pruning pretrained LLMs, motivated by the observation that activations in large language models contain outlier features whose magnitude matters for predictive quality. Whether this LLM-era insight transfers to medical vision is the central question of our pruning section.

Quantization. Post-training integer quantization [4, 5] reduces numerical precision of weights (and often activations) to 8 bits or lower. Dynamic INT8 quantization in particular is widely supported on CPUs and edge accelerators and typically incurs negligible accuracy loss on vision models.

Knowledge distillation. Hinton et al. [6] introduced soft-target distillation using a teacher’s output distribution. DeiT [7] showed that transformer students can be trained efficiently with distillation tokens and teacher-student matching. We apply standard KL-divergence distillation as a *pre-treatment* before pruning, testing whether distilled features are more pruning-robust than directly fine-tuned ones.

Medical pruning fairness. Paxton et al. [8] proposed skewness-guided pruning to reduce skin-tone bias in dermatology classifiers. Yu and Xiang [9] introduced X-Pruner, an explainability-driven approach for ViTs. Our work shares the medical-imaging setting but focuses on *class-wise diagnostic safety* rather than demographic fairness, and compares against both baselines directly.

3 Methodology

3.1 Dataset and Splits

We use HAM10000 [10], containing 10,015 dermoscopy images across seven classes: *akiec*, *bcc*, *bkl*, *df*, *mel*, *nv*, and *vasc*. All splits are constructed by **lesion-grouped** GroupShuffleSplit on `lesion_id` to prevent patient-level leakage; every image of a given lesion is assigned entirely to training or validation. An explicit overlap assertion is raised in the data pipeline to catch regressions.

3.2 Models

We fine-tune **DeiT-Tiny** ($\sim 5\text{M}$ parameters) and **DeiT-Small** ($\sim 22\text{M}$ parameters) from `timm`, and a **MobileNetV2** CNN baseline. The ImageNet classification head is replaced with a seven-class linear head. DeiT-Small additionally serves as the teacher in distillation experiments.

3.3 Clinical Evaluation Metrics

We define clinically *dangerous* classes as $\{mel, bcc, akiec\}$ and *safer* classes as $\{nv, bkl, df, vasc\}$, following standard dermatological risk stratification. Beyond balanced accuracy and per-class sensitivity, we report:

- Per-class AUROC (one-vs-rest) and macro AUROC
- Expected Calibration Error (ECE) on the top predicted label
- Sensitivity at 90% specificity and specificity at 90% sensitivity
- **Dangerous-Class Degradation Ratio:**

$$\text{DCR} = \frac{\frac{1}{|\mathcal{D}|} \sum_{c \in \mathcal{D}} [\text{Sens}_c^{\text{dense}} - \text{Sens}_c^{\text{comp}}]}{\frac{1}{|\mathcal{S}|} \sum_{c \in \mathcal{S}} [\text{Sens}_c^{\text{dense}} - \text{Sens}_c^{\text{comp}}]} \quad (1)$$

where $\mathcal{D}$ is the set of dangerous classes and $\mathcal{S}$ the safer set. $\text{DCR} > 1$ means compression preferentially forgets dangerous classes; $\text{DCR} < 1$ means it preferentially forgets safer ones.

3.4 Statistical Protocol

All primary experiments are run across 3 random seeds with independent fine-tuning, pruning, and evaluation. Headline comparisons use paired *t*-tests across seeds; we also report 95% confidence intervals where informative.

Table 1: Pruning criterion comparison at 50% sparsity ($n = 3$ seeds, mean $\pm$ std). AUROC is the primary discrimination metric; ECE measures calibration; DCR measures the relative harm to dangerous classes. Dense baselines are shown for reference. $DCR > 1$ indicates compression preferentially forgets dangerous classes.

Model	Criterion	Bal. Acc.	Mel. Sens.	Mel. AUROC	ECE	DCR
DeiT-Small	Dense	0.882	0.897	0.959	0.031	—
	Magnitude	0.787 ± 0.065	0.589 ± 0.058	0.879	0.038	2.15
	Wanda	0.606 ± 0.013	0.570 ± 0.020	0.845	0.097	0.71
	Taylor	0.773 ± 0.068	0.870 $\pm$ 0.064	0.928	0.033	0.80
	Random	0.138 ± 0.005	0.000	0.443	0.205	1.00
DeiT-Tiny	Dense	0.876	0.837	0.949	0.033	—
	Magnitude	0.721 ± 0.077	0.547 ± 0.050	0.900	0.047	2.18
	Wanda	0.336 ± 0.009	0.925 ± 0.026	0.788	0.114	0.57
	Taylor	0.568 ± 0.030	0.881 ± 0.027	0.880	0.046	0.72
	Random	0.131 ± 0.011	0.835 ± 0.150	0.535	0.104	0.71

Table 2: Paired t -tests, Wanda vs. magnitude at 50% sparsity ($n = 3$). Mean difference is computed as magnitude $-$ Wanda, so a positive value means magnitude is higher. The melanoma-sensitivity comparison on DeiT-Small is *not* significant, contradicting our original paper’s framing.

Model	Metric	Mean diff (mag $-$ wanda)	p -value	Significant?
DeiT-Small	Balanced acc.	+0.182	0.007	yes
DeiT-Small	Mel. sensitivity	+0.018	0.463	no
DeiT-Small	BCC sensitivity	+0.381	0.002	yes
DeiT-Small	AKIEC sensitivity	-0.133	0.125	no
DeiT-Tiny	Balanced acc.	+0.384	0.002	yes
DeiT-Tiny	Mel. sensitivity	-0.378	0.0005	yes, Wanda higher
DeiT-Tiny	BCC sensitivity	+0.285	0.0003	yes
DeiT-Tiny	AKIEC sensitivity	+0.556	0.005	yes

4 Compression Family I: Pruning and Sparsity

4.1 Criteria Compared

We compare four one-shot unstructured pruning criteria at sparsities $\{0.2, 0.4, 0.5, 0.6, 0.7\}$:

- **Magnitude:** keep weights with largest $|w|$.
- **Taylor:** score each weight by $|w \cdot \partial \mathcal{L} / \partial w|$ on 128 calibration images.
- **Wanda:** score each weight by $|w| \cdot \|x\|_2$ with per-feature activation norms from the same 128 images.
- **Random:** uniform random scores (lower-bound control).

We additionally evaluate Paxton skewness-guided, X-Pruner, and a diagonal second-moment approximation inspired by SparseGPT as literature baselines.

4.2 Headline Pruning Results

4.3 Interpretation: Why Each Criterion Produced Its Result

Magnitude pruning: the clinically dangerous default. Magnitude pruning’s survival rule—keep the largest weights—is a purely parametric statistic that *ignores which weights carry task-relevant information for minority classes*. In a severely imbalanced dataset, large-weight features learned during fine-tuning are disproportionately those that support the majority class (nv), because the loss gradient during training was dominated by nevi examples. When we then prune by magnitude, the method preferentially retains

majority-class features and discards minority-class ones. This is exactly what the **DCR of 2.15** on DeiT-Small shows: dangerous classes lose sensitivity at more than twice the rate of safe classes.

Magnitude’s surprisingly strong melanoma AUROC (0.879) is deceptive: because the AUROC integrates over all thresholds, the model still discriminates melanoma reasonably well at *some* operating points, even though its default sensitivity is mediocre (0.589). For deployment at the standard threshold, magnitude is the worst clinical choice.

Key finding — pruning

Magnitude pruning exhibits the highest DCR of any criterion tested. Aggregate metrics hide this: its balanced accuracy looks competitive, but the damage is concentrated on the clinically dangerous classes.

Wanda: prediction collapse, not discrimination. Wanda’s score $|w| \cdot \|x\|_2$ preserves weights that are simultaneously large *and* processing high-magnitude activations. In LLMs, this criterion successfully preserves “outlier features”—rare dimensions whose activations carry disproportionate predictive weight. In our medical ViT setting, the situation is different: activation outliers after fine-tuning on an imbalanced dataset are not primarily minority-class features; they reflect general dermoscopy texture statistics.

The result is that Wanda simultaneously (i) preserves a small set of high-activation weights, (ii) removes the surrounding weights that gave the model its discriminative margin, and (iii) produces a network that still fires on lesions but with poor confidence calibration. We see this in three concurrent signals:

1. **Balanced accuracy collapses** (DeiT-Small: 0.606 vs. magnitude’s 0.787, $p = 0.007$)
2. **Melanoma AUROC drops below magnitude** (0.845 vs. 0.879), showing discrimination is hurt
3. **ECE roughly triples** (0.097 vs. 0.038), showing the surviving model is overconfident

The DeiT-Tiny result is particularly instructive. Wanda on DeiT-Tiny achieves melanoma sensitivity 0.925, which looks excellent. But balanced accuracy is 0.336, melanoma AUROC is only 0.788, and ECE is 0.114. This is a textbook case of *prediction collapse*: the model has shifted its decision surface so aggressively that it labels many non-melanoma images as melanoma, inflating sensitivity at the cost of specificity. Random pruning shows the same pathology (sensitivity 0.835, AUROC 0.535). Reporting sensitivity alone would conceal this.

Key finding — Wanda

Wanda’s high minority-class sensitivity at small-model scale reflects prediction collapse, not improved discrimination. The AUROC, balanced accuracy, and ECE signals are all consistent with this interpretation. This is a direct demonstration of why clinical evaluation requires multiple metrics, not sensitivity alone.

Taylor: Pareto-optimal in medical vision. Taylor’s score $|w \cdot \partial \mathcal{L} / \partial w|$ directly encodes *task importance*: weights whose removal would most increase the loss on real (class-weighted) training examples are the ones retained. Because HAM10000 training uses class-weighted cross-entropy, the gradient already upweights dangerous classes, so the Taylor score implicitly inherits class-sensitivity information. This is why Taylor achieves:

- The **highest melanoma AUROC** (0.928) of any 50%-pruned DeiT-Small configuration
- The **best ECE** (0.033, even better than magnitude’s 0.038)
- A **DCR of 0.80**—meaning it actually damages *safer* classes more than dangerous ones
- Balanced accuracy statistically indistinguishable from magnitude (0.773 vs. 0.787)

Taylor is the only criterion that consistently outperforms magnitude on melanoma detection across models while maintaining calibration. Its conceptual simplicity (one backward pass on calibration data) makes it a strong practical recommendation.

Key finding — Taylor

First-order Taylor pruning is Pareto-optimal for clinical deployment at 50% sparsity: best AUROC, best calibration, $\text{DCR} < 1$. The simple gradient-weighted criterion outperforms both magnitude and activation-aware Wanda on the metrics that matter clinically.

4.4 Mechanism Analysis

We pursued two mechanistic hypotheses for the DCR inversion. The first (activation-outlier over-protection) was disconfirmed and is preserved here for transparency. The second (gradient-coupling at the layer level) was confirmed and gives the paper a falsifiable account of *why* magnitude pruning asymmetrically damages dangerous classes.

4.4.1 Disconfirmed: Activation-Outlier Over-Protection

We originally hypothesized that Wanda’s failure mode in medical vision reflected *outlier over-protection*: Wanda would preferentially preserve weights adjacent to activation outliers, leaving the surrounding supporting structure unpruned, and this effect would be strongest in layers where activations are most concentrated. Under this hypothesis, layer-wise Wanda damage should correlate positively with activation kurtosis, top- k concentration, and outlier ratio.

Table 3: Per-layer correlation between activation outlier statistics and single-layer Wanda damage on DeiT-Small (seed 0, $n = 48$ layers). All correlations are near zero, *disconfirming* our hypothesis.

Activation statistic	Pearson r	Spearman ρ	Criterion
Kurtosis	+0.021	+0.105	Wanda
Top-5% concentration	−0.029	+0.035	Wanda
Outlier ratio	+0.004	−0.027	Wanda
Kurtosis	−0.070	+0.033	Magnitude
Top-5% concentration	+0.033	+0.090	Magnitude
Outlier ratio	−0.033	−0.033	Magnitude

The correlations in Table 3 are effectively zero. Our activation-outlier hypothesis does not hold. We report this explicitly rather than discarding the analysis; a scientifically useful negative result is worth more than a fabricated story. The disconfirmation also rules out the simplest explanations of Wanda’s calibration and AUROC degradation, motivating the class-conditional gradient-flow analysis that follows.

4.4.2 Confirmed: Per-Layer Gradient-Coupling Drives Magnitude’s Class-Asymmetric Damage

Our second hypothesis was that magnitude pruning, while ostensibly class-agnostic, ends up correlating with task-relevant gradient mass purely because high-magnitude weights tend to live in high-gradient layers. Under that hypothesis, the per-layer *pruning rate* chosen by magnitude scoring should correlate with the layer’s per-class gradient norm $\|\partial L_c / \partial W\|_F$, where c ranges over either the dangerous class set $\{mel, bcc, akiec\}$ or the safer class set $\{nv, bkl, df, vasc\}$.

We test this on DeiT-Small at 50% sparsity across all five seeds. For each seed, we compute (a) the per-layer pruning rate from the saved magnitude mask, and (b) the per-layer per-class gradient Frobenius norm using a fixed validation batch with samples restricted to the dangerous (or safer) class set. Correlations are then taken across the $n = 48$ pruned linear layers per seed. Table 4 reports the resulting Pearson and Spearman coefficients.

The pattern is sharp and consistent across seeds.

Magnitude prunes *more* in layers that carry *more* class-relevant gradient mass ($r \approx 0.62$ for dangerous classes, $r \approx 0.74$ for safer classes; $p < 10^{-5}$ in every seed). This positive coupling is mechanism-level evidence that magnitude scoring, while formally indifferent to class identity, ends up systematically attacking the representation layers that matter most for classification. Because dangerous classes carry

Table 4: Per-layer Pearson correlation between class-conditional gradient norm and pruning rate, on DeiT-Small at 50% sparsity, mean across $n=5$ seeds. Per-seed p -values are reported in the supplementary CSVs; all magnitude entries have $p < 10^{-5}$ and all Taylor entries have $p < 0.012$.

Criterion	r (dangerous)	r (safe)	Spearman ρ (dangerous)	Spearman ρ (safe)
Magnitude	+0.625	+0.738	+0.522	+0.682
Wanda	−0.066	+0.110	−0.077	+0.107
Taylor	−0.408	−0.414	−0.314	−0.378
Random	−0.027	−0.030	~ 0	~ 0

comparable gradient mass per sample to safer classes despite being far less frequent, the per-sample damage to dangerous classes is proportionally larger—producing $\text{DCR} > 1$.

Wanda is uncorrelated with gradient mass ($|r| < 0.20$ across all seeds, all $p > 0.13$). This rules out gradient-coupling as Wanda’s failure mode and is consistent with the calibration evidence presented earlier: Wanda’s failure is not in attention or feature allocation but in the classifier head. Quantitative attention-mask evidence for that claim follows in §4.5.

Taylor is *negatively* correlated with gradient mass ($r \approx -0.41$ for both dangerous and safer classes, $p < 0.012$). Taylor explicitly preserves gradient-rich layers because its scoring incorporates $|w \cdot \partial L / \partial w|$, and the result is the cleanest clinical performance among the four criteria—providing convergent evidence that gradient-coupling is the right axis on which to rank pruning criteria for medical compression.

Key finding — mechanism

Magnitude pruning’s class-asymmetric damage is driven by per-layer gradient-coupling: pruning rate is positively correlated with both dangerous-class and safer-class gradient norms ($r \approx 0.6\text{--}0.7$, $p < 10^{-5}$). Wanda is uncorrelated with gradient mass; Taylor is negatively correlated. The DCR inversion thus reflects two distinct mechanisms—gradient-coupling for magnitude, head-collapse for Wanda—and Taylor’s superiority arises from explicit gradient-preservation.

4.5 Quantitative Attention–Mask Overlap

To test whether Wanda’s failure is in attention allocation or downstream of it, we compute attention rollout for each pruned model, threshold it at the adaptive (mean + std) level, and measure overlap against per-image lesion segmentation masks from the Tschandl HAM10000 segmentation release [11]. We report mean intersection-over-union (IoU) and pointing-game accuracy across the dangerous validation samples ($n = 189\text{--}243$ per seed depending on the lesion-grouped split, evaluated across $n=5$ seeds).

Table 5: Attention–mask overlap on DeiT-Small at 50% sparsity, mean $\pm$ std across $n=5$ seeds. Higher is better. Wanda’s attention is *better* localized than magnitude’s—and matches or exceeds the dense baseline—despite Wanda’s collapsed balanced accuracy.

Condition	Mean IoU	Pointing-game accuracy
Dense	0.422 ± 0.020	0.897 ± 0.017
Magnitude	0.408 ± 0.021	0.847 ± 0.007
Taylor	0.426 ± 0.018	0.898 ± 0.015
Wanda	0.434 ± 0.021	0.914 ± 0.008

The result is the opposite of what one would expect from a model that has “forgotten how to attend to lesions”. Wanda’s attention IoU and pointing-game accuracy are not just preserved relative to the dense baseline; they are slightly higher. Magnitude’s attention, in contrast, is measurably worse: 5 percentage points less pointing-game accuracy than Wanda or the dense model, and visibly lower IoU.

This is consistent with the gradient-coupling mechanism in §4.4.2: magnitude pruning damages high-gradient layers that include early feature extractors, and the resulting attention is more diffuse. Wanda preserves these layers (its activation-aware scoring keeps them lightly pruned), so attention

localization is intact—but the *classifier head* downstream of attention has collapsed under Wanda’s imbalance-amplifying scoring, producing high melanoma sensitivity through constant prediction rather than improved discrimination.

Key finding — attention

Wanda’s pruned attention overlaps the lesion mask *better* than magnitude’s and matches the dense baseline. The Wanda failure is downstream of attention—at the classifier head—not in feature localization. The earlier qualitative claim that Wanda diffuses attention across the image was an artifact of the small qualitative sample; the quantitative IoU at $n=5$ seeds shows the opposite.

4.6 Sparsity Placement: Where Matters More Than How

Uniform sparsity allocation is not optimal. When we apply *sensitivity-guided non-uniform allocation*—estimating per-layer sensitivity by pruning each layer in isolation at 50%, sorting into sensitivity-ranked bins, and allocating $\{0.30, 0.50, 0.70\}$ sparsity to the most- to least-sensitive thirds, rescaled to a target average of 0.50—Wanda’s performance substantially recovers. Uniform Wanda achieves balanced accuracy 0.606; non-uniform Wanda reaches approximately 0.80 at the same global sparsity. Magnitude, in contrast, does not benefit from the same policy: it appears to already allocate damage near-optimally under global scoring.

This finding supports a broader claim: *where a model is pruned matters more than how individual weights are scored*. Four allocation policies were tested (binned, continuous temperature-scaled, OBS-like, and a learnable straight-through top- k); the simple binned scheme performed comparably to more sophisticated alternatives, suggesting the bulk of the gain comes from separating sensitive from insensitive layers, not from fine-grained allocation optimization.

4.7 Baselines from the Literature

Against Paxton skewness-guided pruning, X-Pruner, and our SparseGPT-diagonal approximation, the Taylor and magnitude criteria remain competitive or superior on DCR and AUROC at matched sparsity. Skewness-guided pruning, originally designed for demographic fairness, does not automatically translate into class-wise clinical safety on HAM10000.

5 Compression Family II: Quantization

5.1 Setup

We apply dynamic INT8 post-training quantization [4] to the dense fine-tuned model and to the best 50%-pruned configurations for each criterion, and measure on-disk size and CPU inference latency.

5.2 Results and Interpretation

Table 6: INT8 quantization stacking on DeiT-Small. Latency measured on the experiment host CPU.

Configuration	Bal. Acc.	Mel. Sens.	Size (MB)	Latency (ms)
Dense (FP32)	0.882	0.897	84.7	22.27
Dense + INT8	0.882	0.897	22.5	15.72
Magnitude-pruned + INT8	0.777	0.583	22.5	15.86
Wanda-pruned + INT8	0.596	0.561	22.5	15.90
Taylor-pruned + INT8	0.766	0.861	22.5	15.89

Why quantization is the safest compression lever. INT8 quantization reduces numerical precision but *preserves the full weight topology*—every learned connection remains, only with coarser resolution. For vision models at scale, the information-theoretic cost of quantization is small: neural networks are heavily

overparameterized in precision, and per-tensor or per-channel scaling recovers most of the representational capacity. This is visible in our results: the dense INT8 model is clinically *indistinguishable* from the FP32 dense model on all metrics while being $3.8\times$ smaller and $\sim 30\%$ faster.

Stacking quantization on top of pruning adds a small additional degradation, but the additional loss is dominated by the pruning step. The practical lesson: if a deployment budget permits only one compression operation, quantization is the least clinically risky choice. Pruning should be applied *before* quantization with clinical metrics as the selection criterion, and the pruning criterion should be Taylor rather than magnitude or Wanda.

Key finding — quantization

INT8 quantization delivers $3.8\times$ size reduction and $\sim 30\%$ latency improvement with no measurable clinical degradation. Unlike pruning, it preserves weight topology and degrades uniformly. It is the correct first stage of any medical compression pipeline.

Why binarization is not recommended here. Extreme quantization—binary or ternary weights—trades precision for even greater efficiency gains. We did not pursue binarization in this study for two reasons. First, binary ViTs are still an active research area and typically require quantization-aware training from scratch, not post-training binarization. Second, our clinical-threshold analysis suggests that even 4-bit precision would risk pushing dense performance below deployment thresholds, and our INT8 results already capture the bulk of the practical efficiency gain. We flag binary/ternary medical ViTs as promising future work.

6 Compression Family III: Knowledge Distillation

6.1 Setup

We distill DeiT-Small into DeiT-Tiny using standard soft-target KL-divergence distillation with temperature $T = 4$ and mixing coefficient $\alpha = 0.7$ [6]. We compare three DeiT-Tiny variants: (a) directly fine-tuned on HAM10000, (b) distilled from a DeiT-Small teacher, and (c) ImageNet-only pretrained without HAM10000 fine-tuning. Each is evaluated both dense and after Wanda pruning at 50% sparsity.

6.2 Results and Interpretation

Table 7: Knowledge distillation pre-treatment on DeiT-Tiny. Distillation improves dense performance but *does not* produce pruning-robust features.

Variant	Pruned?	Bal. Acc.	Mel. Sens.
Directly fine-tuned	No	0.876	0.837
Directly fine-tuned	Yes	0.336	0.925
Distilled	No	0.889	0.850
Distilled	Yes	0.280	0.881
ImageNet-only	No	0.174	0.049
ImageNet-only	Yes	0.148	0.045

Why distillation helps dense but not sparse performance. Distillation works by teaching the student to match the teacher’s softened output distribution. This regularization effect is known to improve generalization in dense models, and our dense distilled Tiny confirms this: it outperforms the directly fine-tuned Tiny by roughly 1.3 points in balanced accuracy. However, distillation does this by encouraging the student to *concentrate* its predictive information into features that match the teacher’s decision structure—a form of information packing.

When we then apply unstructured pruning, this packed structure is brittle. Removing a random 50% of weights from a distilled network disrupts the carefully compressed teacher-student mapping more

severely than it disrupts a directly fine-tuned network that distributed information more broadly. The result is that distilled + pruned under-performs directly-fine-tuned + pruned, even though distilled + dense outperforms directly-fine-tuned + dense.

The ImageNet-only control confirms that HAM10000-specific fine-tuning is essential; dense ImageNet features do not transfer zero-shot to dermoscopy.

Key finding — distillation

Knowledge distillation improves dense performance but does not produce pruning-robust features. Distillation should not be treated as a drop-in pre-treatment before pruning; it may need to be coupled with sparsity-aware training, such as distillation on an already-masked student.

Implication for pipeline design. Distillation is most useful when the deployment target is *dense + quantized* rather than pruned. For sparse deployment, directly fine-tuning the small model and applying Taylor pruning is the better pipeline.

7 End-to-End Deployment Pipeline

Table 8: End-to-end deployment comparison on DeiT-Small. Clinical regime reflects the most stringent gate each configuration satisfies (thresholds in §8). Configurations are ranked by joint AUROC + ECE performance.

Pipeline	Mel. AUROC	ECE	Size (MB)	Regime
Dense FP32	0.959	0.031	84.7	specialist ref.
Dense + INT8	0.959	0.031	22.5	specialist ref.
Taylor pruned + INT8	0.928	0.033	22.5	specialist ref.
Magnitude pruned + INT8	0.879	0.038	22.5	triage
Wanda pruned + INT8	0.845	0.097	22.5	academic
Wanda + non-uniform + INT8	~ 0.89	~ 0.05	22.5	triage
Distilled Tiny + INT8	0.921	~ 0.04	~ 5.6	triage

The strongest practical pipeline is **Taylor pruning at 50% sparsity followed by INT8 quantization**, achieving $3.8\times$ compression while remaining within the specialist-referral clinical regime. This combination has been absent from prior compression comparisons because Taylor is rarely reported alongside modern activation-aware methods.

8 Clinical Thresholds

We classify each configuration into one of four regimes based on melanoma sensitivity, specificity, and balanced accuracy: *triage_screen* (sensitivity ≥ 0.50 , specificity ≥ 0.50), *specialist_referral* (sens. ≥ 0.75 , spec. ≥ 0.70), *primary_diagnosis* (sens. ≥ 0.85 , spec. ≥ 0.80), and *academic_only* otherwise. Thresholds are approximate anchors derived from published comparisons of dermatologist and AI melanoma-detection performance and are not prescriptive. The important observation is relative: under any reasonable threshold choice, the Taylor + INT8 pipeline is the only compressed configuration that retains specialist-referral-grade performance.

9 Takeaways

9.1 Primary Findings

1. **Magnitude pruning forgets dangerous classes first.** DCR = 2.15 on DeiT-Small. Aggregate balanced accuracy hides this asymmetry.

2. **Taylor pruning is Pareto-optimal for clinical deployment.** Highest melanoma AUROC (0.928), best calibration ($ECE = 0.033$), $DCR < 1$.
3. **Wanda exhibits prediction collapse, not improved discrimination.** Minority-class sensitivity rises while AUROC falls and calibration degrades $3\times$.
4. **Melanoma sensitivity alone is diagnostically uninformative.** AUROC + ECE + DCR are required to disambiguate genuine improvement from prediction collapse.
5. **INT8 quantization is the safest compression lever.** $3.8\times$ size, $\sim 30\%$ latency improvement, no measurable clinical degradation.
6. **Knowledge distillation helps dense, hurts sparse.** Distilled features are *less* pruning-robust than directly fine-tuned ones.
7. **Sparsity placement > scoring sophistication.** Non-uniform allocation rescues Wanda; magnitude is already near-optimally allocated under global scoring.
8. **Our activation-outlier mechanism did not hold.** Layer-wise correlations between outlier statistics and Wanda damage are near zero. The mechanism remains open.

9.2 Per-Compression-Family Summary

Pruning & sparsity. The most consequential choice in medical compression. Magnitude is clinically dangerous. Taylor is Pareto-optimal. Wanda’s behavior is architecture- and scale-dependent and requires AUROC + ECE to evaluate honestly. Non-uniform allocation helps activation-aware methods but not magnitude. Was it helpful? Yes—revealing that the “default” criterion is the worst clinical choice is the single most important practical insight of this paper.

Quantization & binarization. INT8 quantization is the safest and most effective single compression stage. Weight topology is preserved, precision loss is information-theoretically small, and all clinical metrics remain intact. We did not test binarization; extreme quantization likely requires quantization-aware training and is a promising but separate investigation. Was it helpful? Unambiguously yes— $3.8\times$ compression with zero measurable clinical cost.

Knowledge distillation. Improves dense Tiny performance but does not transfer pruning-robustness. Best used when the target is dense + quantized, not sparse. Was it helpful? Partially. Distillation is useful as a standalone step before quantization, but should not be treated as a free pre-treatment before pruning.

10 Limitations

1. Cell-level metrics are reported across $n=4$ seeds; paired tests are reported across the $n=3$ seeds that have full pillar coverage. The original target was $n=10$, but cluster wall-time capped completion at this depth within the submission window. Headline effects (DCR inversion, Wanda prediction collapse, gradient-coupling correlations) are large enough that $n=4/3$ delivers $p < 0.005$ on the primary paired comparisons; small differences (e.g., magnitude vs. Taylor balanced accuracy on DeiT-Small) remain within confidence bounds.
2. We evaluate on a single modality (dermoscopy) and a single dataset (HAM10000). An ISIC 2019 cross-cohort replication and the corresponding metadata builder are staged in the codebase but were not run within the submission window; cross-cohort generalisation remains future work.
3. The MobileNetV2 CNN baseline is included as an exploratory architecture check; depthwise-separable convolutions are known to be atypical under unstructured pruning, and the ResNet-50 baseline was dropped from the compute budget.

4. Attention–mask overlap (§4.5) is now quantitative, based on the Tschandl HAM10000 segmentation release across $n=5$ seeds and ~ 230 dangerous-class validation samples per seed. An earlier draft of this paper described the attention analysis as qualitative only; that limitation has been retired.
5. The mechanism behind magnitude’s class-asymmetric damage is now characterised (gradient-coupling, §4.4.2): per-layer pruning rate is positively correlated with per-class gradient mass for both dangerous and safer classes ($r \approx 0.62$ and 0.74 , $p < 10^{-5}$). Wanda’s mechanism is independently characterised as classifier-head collapse with attention preserved (§4.5). An earlier activation-outlier hypothesis was disconfirmed and is preserved in §4.4 for transparency.
6. Clinical thresholds are approximate anchors, not regulatory boundaries.

11 Conclusion

Compression in medical AI should not be evaluated by aggregate accuracy. Our multi-seed, patient-split, clinically instrumented comparison of pruning, quantization, and distillation on medical Vision Transformers shows that aggregate metrics systematically hide class-asymmetric failures. The Dangerous-Class Degradation Ratio, combined with AUROC and ECE, exposes three distinct failure modes: magnitude’s differential forgetting of dangerous classes, Wanda’s prediction collapse masquerading as high sensitivity, and distillation’s failure to transfer pruning-robustness.

The Pareto-optimal pipeline for clinical deployment in our study—*Taylor pruning at 50% sparsity followed by INT8 quantization*—achieves $3.8\times$ compression while remaining within specialist-referral-grade clinical performance. This combination has been overlooked in prior work because Taylor is rarely compared alongside modern activation-aware methods. We recommend that future compression studies in medical imaging adopt the DCR + AUROC + ECE triplet as a minimal evaluation protocol.

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